

Python Practice Questions

Topics: Tuple, Nested Tuple, Dictionary, Nested Dictionary

1. Smart Attendance Analyzer

- A company stores employee attendance in nested dictionary format.
- Find employee with highest attendance percentage.
- Find employees absent consecutively for 2 days.
- Create a new dictionary containing only employees with attendance greater than 70 percent.

2. AI Model Performance Tracker

- Store ML model details using nested dictionaries.
- Find best model using highest accuracy and lowest loss.
- Create ranking manually without using sort().
- Detect unstable models where loss is greater than 0.20.

3. Multi-City Weather Monitoring System

- Nested dictionary stores weather data of multiple cities.
- Find hottest city using average temperature.
- Find city with highest temperature fluctuation.
- Create alert dictionary for cities crossing 42 degrees.

4. Fraud Transaction Detector

- Transactions are stored as nested tuples.
- Detect suspicious users whose total transaction is greater than 100000.
- Detect users having more than one high-value transaction.
- Generate blacklist users list.

5. E-Commerce Recommendation Engine

- Store user purchase history using dictionary and list.
- Recommend products user has not purchased but similar users purchased.
- Find most common product.
- Create product popularity ranking manually.

6. Hospital Emergency Prioritizer

- Patient records are stored in nested dictionaries.
- Create emergency priority levels such as Critical, Moderate and Normal.
- Find patient needing immediate attention.
- Generate summary report dictionary.

7. Nested Tuple Matrix Analyzer

- Matrix is stored as nested tuples.

- Find diagonal sum.
- Rotate matrix by 90 degrees manually.
- Flatten matrix without using list comprehension.
- Find prime numbers inside matrix.

8. AI Dataset Cleaner

- Dataset records contain missing and invalid values.
- Remove incomplete rows.
- Replace missing values logically.
- Count valid and invalid records.
- Generate cleaned dataset dictionary.

9. Smart Traffic Monitoring System

- Traffic data is stored in nested dictionary format.
- Find most dangerous city.
- Create traffic risk score formula.
- Rank cities manually using risk score.
- Detect safest city.

10. Cyber Security Login Tracker

- Login attempts are stored in dictionary format.
- Detect suspicious users.
- Block users after 3 consecutive failures.
- Find most stable user.
- Generate final security report dictionary.

Instructions: Solve all questions using loops, conditions, tuples, nested tuples, dictionaries and nested dictionaries only.